REASONING

3. Action with sodium bicarbonate:

Carboxylic acids react with sodium bi carbonate and liberate CO_2. Evolution of carbon dioxide gives brisk effervescence.

$$2R\text{-}COOH + 2NaHCO_3 \longrightarrow 2R\text{-}COONa + CO_2 \uparrow + H_2O$$

4. Action with Borsches reagent:

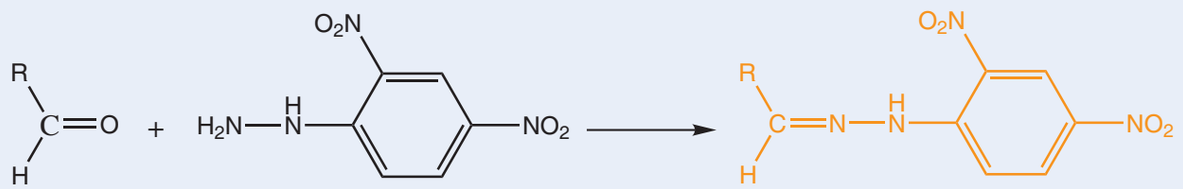
Borsches reagent is prepared by dissolving 2,4-dinitrophenylhydrazine in a solution containing methanol and little of conc sulphuric acid.

Aldehydes and ketones react with borsches reagent to form yellow, orange or red precipitate (dinitro phenylhydrazone)

Aliphatic carbonyl compounds give deep yellow precipitate.

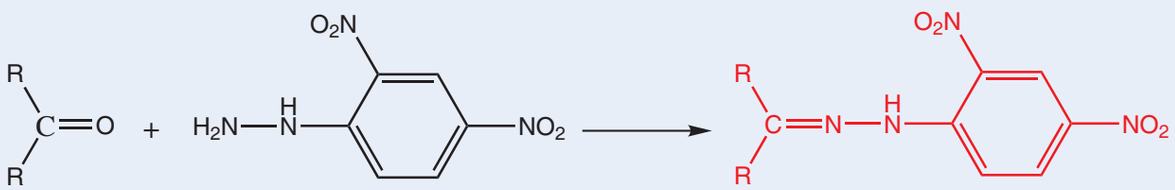
Aromatic carbonyl compounds give red precipitate.

2,4-dinitrophenyl hydrazine can be used to qualitatively detect the carbonyl group of an aldehyde or ketone. A positive result is indicated by the formation of an yellow or orange-red precipitate of 2,4-dinitrophenyl hydrazone.



Aldehyde 2,4 dinitrophenylhydrazine

Aldehyde 2,4 dinitrophenylhydrazone (Yellow or orange)



Ketone 2,4 dinitrophenylhydrazine

Ketone 2,4 dinitrophenylhydrazone (Yellow or orange or red)

5.Charring test:

When carbohydrates are treated with concentrated sulphuric acid, dehydration of carbohydrates results in charring.

$$C_x(H_2O)_y \xrightarrow[\Delta]{H_2SO_4} x\,C + yH_2O$$

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